13 Authentication and Authorization

13.1 Introduction

Authentication and authorization are two critical components in modern web application security. **Authentication** is the process of verifying the identity of users, ensuring they are who they claim to be. On the other hand, **authorization** determines what an authenticated user is allowed to do within the application. Laravel provides a powerful and flexible authentication and authorization system out of the box, simplifying the development of secure web applications. This chapter explores how Laravel 12 supports these features, from basic login systems to advanced role and permission controls.

13.2 Setting Up Authentication with Laravel Breeze

Laravel Breeze is a minimal and simple implementation of Laravel's authentication features. It includes routes, controllers, and views for login, registration, password resets, and more. To get started, ensure that you have a Laravel project set up. Then, install Laravel Breeze using Composer:

```
composer require laravel/breeze --dev
php artisan breeze:install
npm install && npm run dev
php artisan migrate
```

Once the scaffolding is in place, Laravel provides you with a complete authentication system using either Blade or Inertia. This includes user registration, login, logout, email verification (optional), and password resets. These features lay the groundwork for building secure user access systems.

13.3 Understanding the Authentication Flow

Laravel's authentication flow is built on middleware and guards. Middleware such as `auth` ensures that only authenticated users can access certain routes. The `web` middleware group includes session state and CSRF protection, making it suitable for traditional web applications.